

Somu Singh

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EDUCATION

Institute of Engineering & Technology, DAVV | Indore, MP

October 2022 - Present

- Bachelor of Engineering in Computer Engineering | **GPA: 9.02**
- Relevant Coursework: Data Structures & Algorithms, Operating System, Database Management, Computer Programming (C++, Java), Object Oriented Programming, Software Engineering, Web Technologies

Mount Litera Zee School | Indore, MP

- Higher Secondary Education | CBSE | Class XII : **95%** 2020 - 2021
- Secondary Education | CBSE | Class X : **90%** 2018 - 2019

TECHNICAL PROJECTS

Catalyst | AI-Powered Developer Productivity Suite ([Github](#) • [Live](#)) | **Next.js, Gemini, RAG, tRPC, Prisma, Tree-sitter**

- Developed an AI-powered suite of developer tools designed to accelerate code comprehension, project analysis, and technical learning.
- **GitWhiz**: Implemented a Retrieval-Augmented Generation (RAG) pipeline, enabling developers to ask natural language questions about any GitHub repository. Built features for AI-driven commit summaries and interactive codebase analysis to provide deep project insights.
- **Dependency Graph**: Engineered an interactive dependency graph visualizer using ReactFlow and Tree-sitter, parsing source files at the AST level to extract entities, imports, and exports. Rendered file-level relationships as a navigable graph with real-time layout controls, node-level metrics (cyclomatic complexity, LOC, function count), and collapsible sidebars for deep-dive code inspection.

Ashura Core | ESP32 Smart Desk Companion OS ([Github](#)) | **C++, ESP32-S3, WebSockets, Embedded Systems**

- Engineered a custom embedded OS for ESP32-S3 featuring a modular app launcher, stack-based screen navigation, and an pub/sub system for fully decoupled inter-module communication.
- Built a fault-tolerant WebSocket client (retry, heartbeat, message routing) for real-time communication with a chain-of-responsibility MessageRouter to dispatch incoming JSON commands to registered services..
- Developed a distributed device controller with HTTP + mDNS discovery and persistent device registry.

Huddle | Personal Project ([Github](#) • [Live](#)) | **Next.js, Prisma, Clerk, Socket.IO, LiveKit, Zustand, PostgreSQL**

- Built a scalable real-time messaging system supporting servers, channels, and DMs using WebSockets.
- Implemented **RBAC**, message editing, soft deletion, and file uploads.
- Integrated **WebRTC (LiveKit)** for secure video/audio calls with token-based authentication.
- Optimized client-server state using Zustand & React Query for efficient data synchronization.

SKILLS

Programming Languages: Java, SQL, JavaScript, C++, TypeScript, Python

Frameworks & Libraries: React.js, Next.js, Express.js, Node.js, TailwindCss, shadcn, DaisyUI, leaflet.js

Tools and Platforms: GIT, MongoDB, Postman, Prisma, Clerk, Convex

Operating System: Windows, Linux

ACHIEVEMENTS

Top 10 Finalist – HackByte 3.0, IIITDM Jabalpur ([Certificate](#))

Selected among 120+ teams for developing an impactful AI-powered collaboration platform for government departments.

CERTIFICATIONS

Postman API Fundamentals Student Expert ([Certificate](#))

Networking Basics Badge ([Issued by Cisco](#))